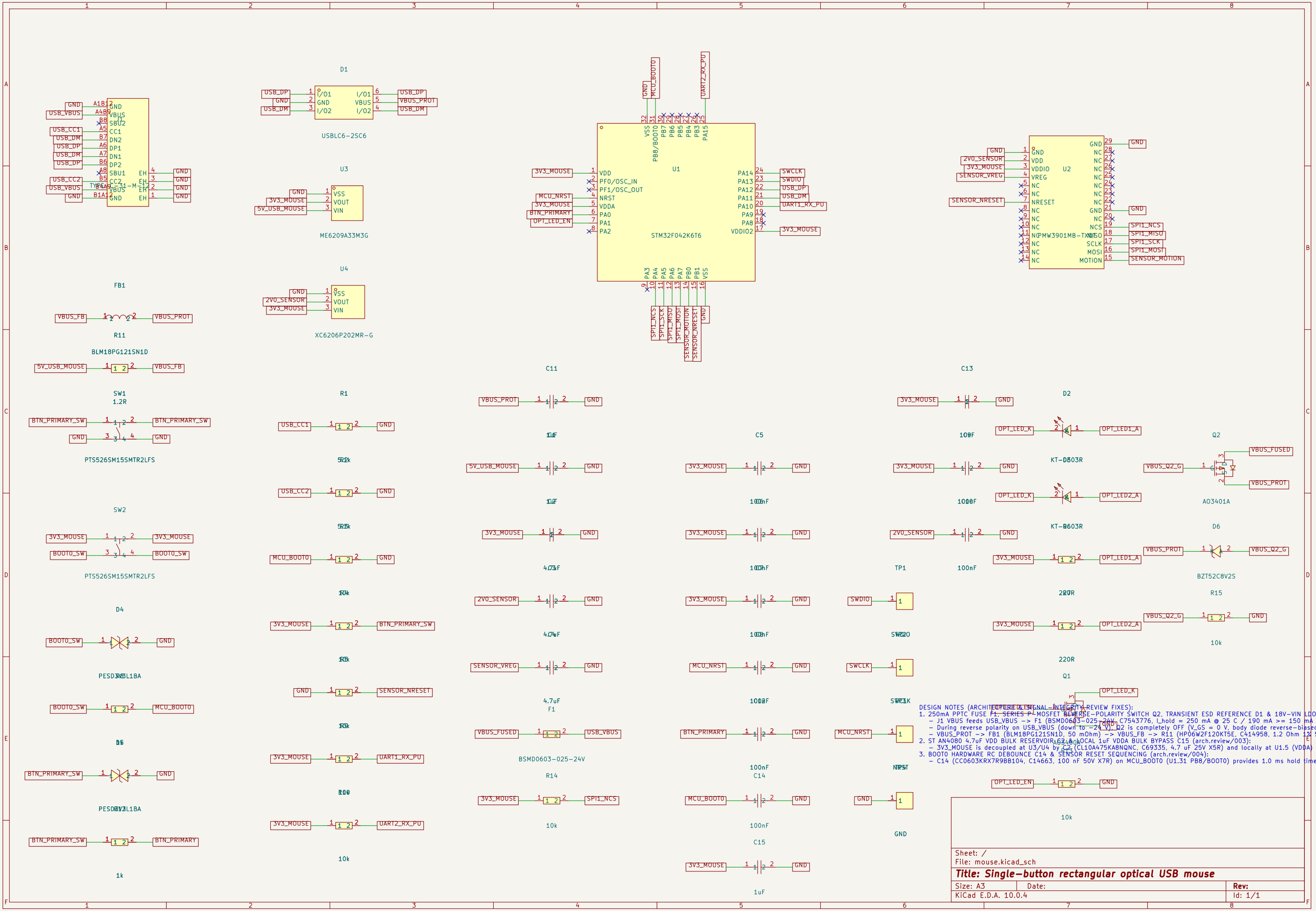

GND
USB_VBUS
USB_CC1
USB_DM
USB_DP
USB_CC2
GND
USB_VBUS
TYPE-C-31-M-12
GND
VBUS
SBU2
CC1
DN2
DP1
DN1
DP2
SBU1
VBUS
GND
EH
EH
EH
EH
GND
GND
GND
GND
D1
USB_DP
GND
USB_DM
I/O1
GND
I/O2
I/O1
VBUS
I/O2
USB_DP
VBUS_PROT
USB_DM
USBLC6-2SC6
U3
GND
3V3_MOUSE
5V_USB_MOUSE
VSS
VOUT
VIN
ME6209A33M3G
U4
GND
2V0_SENSOR
3V3_MOUSE
VSS
VOUT
VIN
XC6206P202MR-G
FB1
VBUS_FB
VBUS_PROT
R11
BLM18PG121SN1D
5V_USB_MOUSE
VBUS_FB
SW1
1.2R
BTN_PRIMARY_SW
BTN_PRIMARY_SW
GND
GND
PTS526SM15SMTR2LFS
SW2
3V3_MOUSE
3V3_MOUSE
BOOT0_SW
BOOT0_SW
PTS526SM15SMTR2LFS
D4
BOOT0_SW
GND
PESD3V3L1BA
BOOT0_SW
MCU_BOOT0
D5
BTN_PRIMARY_SW
GND
PESD3V3L1BA
BTN_PRIMARY_SW
BTN_PRIMARY
1k
R1
USB_CC1
GND
5.1k
USB_CC2
GND
5.1k
MCU_BOOT0
GND
10k
3V3_MOUSE
BTN_PRIMARY_SW
10k
SENSOR_NRESET
GND
10k
3V3_MOUSE
UART1_RX_PU
10k
3V3_MOUSE
UART2_RX_PU
10k
U1
GND
MCU_BOOT0
UART2_RX_PU
VSS
PB8/BOOT0
PB7
PB6
PB5
PB4
PB3
PA15
3V3_MOUSE
VDD
PF0/OSC_IN
PF1/OSC_OUT
NRST
VDDA
PA0
PA1
PA2
MCU_NRST
3V3_MOUSE
BTN_PRIMARY
OPT_LED_EN
STM32F042K6T6
PA14
PA13
PA12
PA11
PA10
PA9
PA8
VDDIO2
SWCLK
SWDIO
USB_DP
USB_DM
UART1_RX_PU
3V3_MOUSE
PA3
PA4
PA5
PA6
PA7
PB0
PB1
VSS
SPI1_NCS
SPI1_SCK
SPI1_MISO
SPI1_MOSI
SENSOR_MOTION
SENSOR_NRESET
GND
C11
VBUS_PROT
GND
1uF
5V_USB_MOUSE
GND
1uF
3V3_MOUSE
GND
4.7uF
2V0_SENSOR
GND
4.7uF
SENSOR_VREG
GND
4.7uF
F1
VBUS_FUSED
USB_VBUS
BSMD0603-025-24V
R14
3V3_MOUSE
SPI1_NCS
10k
C5
3V3_MOUSE
GND
100nF
3V3_MOUSE
GND
100nF
3V3_MOUSE
GND
100nF
3V3_MOUSE
GND
100nF
MCU_NRST
GND
100nF
C14
BTN_PRIMARY
GND
100nF
C15
MCU_BOOT0
GND
100nF
3V3_MOUSE
GND
1uF
U2
GND
2V0_SENSOR
3V3_MOUSE
VREG
SENSOR_NRESET
GND
VDD
VDDIO
NC
NC
NRESET
NC
NC
NC
NCS
MISO
SCLK
MOSI
MOTION
PMW3901MB-TXQT
GND
NC
NC
NC
NC
NC
NC
NC
NC
NC
NC
NC
GND
SPI1_NCS
SPI1_MISO
SPI1_SCK
SPI1_MOSI
SENSOR_MOTION
C13
3V3_MOUSE
GND
100nF
D2
OPT_LED_K
OPT_LED1_A
KT-0603R
OPT_LED_K
OPT_LED2_A
KT-0603R
TP1
2V0_SENSOR
GND
100nF
SWDIO
SWDIO
SWCLK
SWCLK
MCU_NRST
NRST
GND
Q2
VBUS_FUSED
VBUS_Q2_G
VBUS_PROT
AO3401A
D6
VBUS_PROT
VBUS_Q2_G
BZT52C8V2S
R15
3V3_MOUSE
OPT_LED1_A
220R
3V3_MOUSE
OPT_LED2_A
220R
Q1
OPT_LED_K
OPT_LED_EN
GND
2N7002K
10k
VBUS_Q2_G
GND
10k
DESIGN NOTES (ARCHITECTURE & SIGNAL-INTEGRITY REVIEW FIXES):
1. 250mA PPTC FUSE F1, SERIES P-MOSFET REVERSE-POLARITY SWITCH Q2, TRANSIENT ESD REFERENCE D1 & 18V-VIN LDO (arch.re
– J1 VBUS feeds USB_VBUS -> F1 (BSMD0603-025-24V, C7543776, I_hold = 250 mA @ 25 C / 190 mA >= 150 mA @ 50 C,
– During reverse polarity on USB_VBUS (down to -24 V) D2 is completely OFF (V_GS = 0 V, body diode reverse-biased), isolat
– VBUS_PROT -> FB1 (BLM18PG121SN1D, 50 mOhm) -> VBUS_FB -> R11 (HP06W2F120KT5E, C414958, 1.2 Ohm 1/2 500 mW
2. ST AN4080 4.7uF VDD BULK RESERVOIR & 0.1uF LOCAL 1uF VDDA BULK BYPASS C15 (arch.review/003):
– 3V3_MOUSE is decoupled at U3/U4 by C2 (CL10A475KA8NQNC, C69335, 4.7 uF 25V X5R) and locally at U1.5 (VDDA) / U1.1 (
3. BOOT0 HARDWARE RC DEBOUNCE C14 & SENSOR RESET SEQUENCING (arch.review/004):
– C14 (CC0603KRX7R9BB104, C14663, 100 nF 50V X7R) on MCU_BOOT0 (U1.31 PB8/BOOT0) provides 1.0 ms hold time (R3 10k
Sheet: /
File: mouse.kicad_sch
Title: Single-button rectangular optical USB mouse
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/1